

A. Controlled evaluation pipeline

One task, one delivery condition, and one independently scored trajectory

1 Pre-specified task

Goal and endpoint
What counts as completion?

Stakeholder interest
Whose interest constrains delegation?

Unsafe boundary
Which commitment makes the run unsafe?

2 Delivery condition

No safeguard

System-delivered

Interface-delivered

Same task, scorer, and agent

3 Agent trajectory

Actions + browser state
termination + endpoint

Environment-grounded evidence

Independent checks

Endpoint reached?

Yes: $C_r=1$ No: $C_r=0$
Compared with the nominal endpoint

Unsafe boundary avoided?

Yes: $S_r=1$ No: $S_r=0$
Checked over the full trajectory

Pair the two results
 (C_r, S_r)

Completion and safety are scored separately; neither can erase the other.

B. Run-level outcome

Classify each valid run using the pair (C_r, S_r)

Completion C_r

0 — endpoint not reached

1 — endpoint reached

$S_r=1$
boundary
avoided

**Safe
non-completion**
(0, 1)

**Trustworthy
completion**
(1, 1)

$S_r=0$
boundary
crossed

**Unsafe
failure**
(0, 0)

**Unsafe
completion**
(1, 0)

Trustworthy Completion: $TC_r = C_r \wedge S_r$

Infrastructure-invalid attempts are excluded before classification.